

AWS CLI vs. Infrastructure as Code (IaC)

Overview

AWS CLI is a command-line tool for interacting with AWS services. Infrastructure as Code (IaC) is the practice of defining and managing cloud infrastructure using code files rather than manual configuration.

Key Differences

AWS CLI: Executes commands one at a time to manage AWS resources.

IaC: Defines the desired infrastructure in reusable code.

AWS CLI: Imperative approach: tell AWS exactly what to do.

IaC: Declarative approach: describe the desired end state.

AWS CLI: Great for administration, scripting, and automation.

IaC: Great for repeatable deployments, version control, and consistency.

Examples

AWS CLI

`aws s3 ls`

Lists S3 buckets.

`aws ec2 describe-instances`

Lists EC2 instances.

CloudFormation IaC Example

Resources:

MyBucket:

Type: AWS::S3::Bucket

How They Work Together

A common workflow is to define infrastructure using CloudFormation, AWS CDK, or Terraform, and then deploy it using the AWS CLI (for example, `aws cloudformation deploy`).

Recommended Learning Path

1. AWS CLI
2. AWS CloudFormation
3. AWS CDK
4. Terraform

Conclusion

Use the AWS CLI for day-to-day management and scripting. Use IaC to build, version, and consistently deploy cloud infrastructure. Together they form a powerful foundation for cloud engineering and architecture.